CST1500 Coursework One – SOB Portfolio

Deadline for submission: Friday 1st November 2024 18:00 (Week 6).

NOTE: Due to expected high use, there will be a 2-hour window where submissions will be accepted and will not incur a late penalty. Submissions received after 20:00 will be treated as late and will lose 10% of their final mark (to a limit of no lower than 40%, providing the final mark before penalties is above 40%). Submissions received more than 24 hours late will receive zero marks. DO NOT TREAT THIS AS A 20:00 DEADLINE – I advise starting your submission as early as possible before the 18:00 deadline.

Please read all the assignment specification carefully first, before asking your tutor any questions.

General Information

The first piece of coursework for CST1500 is a portfolio of work that you will complete over the first 6 weeks of the course. The portfolio will comprise 10 individual pieces of work that fulfil 10 SOBs (Student Observable Behaviours). The deliverable for each SOB is specified on pages 5 - 9. All 10 SOBs must be submitted to the dedicated assignment link in the Coursework One Submission folder on the module page by the specified deadline.

Your work will not be accepted as an email attachment.

Therefore, you are strongly advised to allow plenty of time to upload your work prior to the deadline.

NOTE: This is an **INDIVIDUAL** Coursework and should **NOT** be completed in pairs or in groups. Please refer to Section 6 of the module handbook with regards to integrity and academic misconduct.

Coursework Weighting

50% of the module.

What to Submit

A zipped folder containing all of the files and folders as specified in the SOBs CW1SOB1 – CW1SOB10.

IMPORTANT: To enable consistency and ease of marking the folder should be named **CST1500CW1** and the filename of each SOB should be in the following format:

CW1SOB1
CW1SOB2
CW1SOB3 etc.

For example: CW1SOB1.c, CW1SOB8.bat, etc.

You may be required to record some video demos, if that is the case, name the video file the same as the SOB to which it pertains i.e. a video demo for CW1SOB4 would be named CW1SOB4.mp4.

The following image is an example of the contents of the **CST1500CW1** folder and is similar to what your submission may look like:

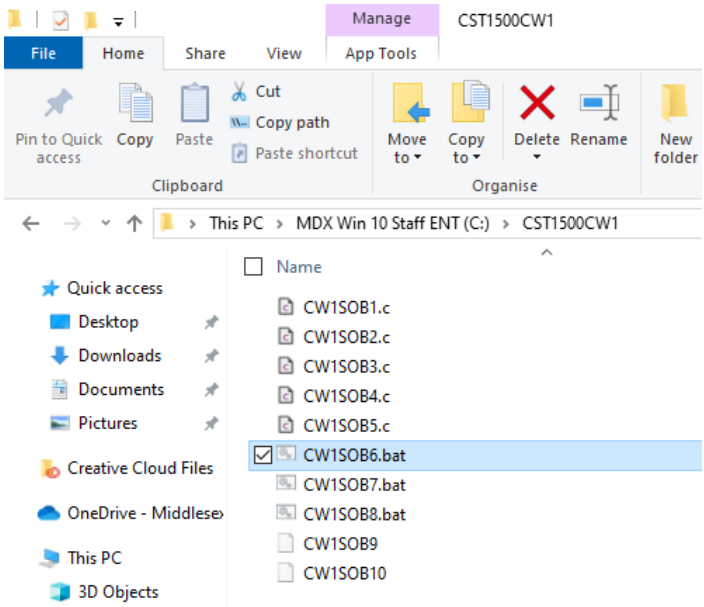

Fig. 1: Example folder contents.

IMPORTANT NOTE: Some SOBs require you to create additional files and folders. These files and folders should also be present in your submission.

Deadline

Friday 1st November 2024 18:00 (Week 6).

Where to submit

Submit your zipped folder to the dedicated submission link in the Coursework One Submission folder on the course page.

Fig. 2: Coursework One submission link on the CST1500 course page.

Demonstration of each SOB

Each SOB must be demonstrated to your lab tutor at a specified time in the lab in week 6 – your lab tutor will notify you of the specific time in due course.

The demo will be to run your programs/scripts and show the output. If you have not completed a SOB, you can show what you have done thus far.

You can also demonstrate the SOBs earlier during the labs if there is adequate time and a lab tutor is available. A record will be kept of students who have demonstrated SOBs at an earlier time.

IMPORTANT

If you cannot demonstrate some/all of the SOBs in your lab, you must make separate video demos for each SOB not demonstrated in your lab.

Details of how to record a video demonstration can be found on the course page in the Coursework One section.

Therefore, you are strongly advised to attend the specified lab time to demonstrate your work.

IMPORTANT

If you do not demonstrate your work either in the lab or via a submitted video, 1 mark will be deducted from each SOB not demonstrated.

Marking

This piece of coursework is worth **50%** of the overall mark for CST1500. Each of the 10 SOBs carry a weighting of **5%**.

Each **5%** for each SOB is distributed and applied using the following relative marking criteria:

Criterion	Marks
Correctness of output	2
Code/script formatting and sensible use of comments in code	1
Following instructions	1
Demo of work – in class or video	1

Half Marks will be awarded where appropriate.

All marks are provisional and subject to adjustment during moderation of the overall module assessment which takes place at the end of the year with external examiner oversight.

Use of AI

AI is a powerful tool that can increase the productivity of programmers. You are welcome to use AI tools like GitHub Copilot, ChatGPT and CodeWhisperer to help you to write code for your coursework. You are also welcome to use text generation tools like ChatGPT and Microsoft Copilot to help write your reports.

AI tools can help you to program and write more efficiently, but they are not going to complete the coursework for you, they frequently make mistakes, and they are unlikely to follow the technology requirements in the coursework descriptions. If you use these tools to generate the code and report for you without understanding the technologies or coursework requirements, then you will get a low mark for this module.

I strongly recommend that you spend time understanding how the programming languages (C, Bash, Batch, Python etc.) work and write code with these technologies without the help of AI. Then you will be in a much better position to use AI tools to help you to generate drafts of pieces of code for your project. Your understanding of the technologies will enable you to change this code to match the project requirements, integrate it, and fix bugs.

Detailed specification of each SOB

CW1SOB1 – A simple C program

Write a **C program** that displays your name, student ID, Degree Programme and why you are studying that programme.

The program should also perform 4 calculations using the numbers in your student ID (+, -, /, and *).

Include some relevant comments in your code.

CW1SOB2 – A C program with user input

Write a **C program** that takes some numbers entered by the user and performs some operations on them (+, -, /, and *).

The programme should also take the name of the user and display it in every result e.g.

```
Student_Name has entered X and Y. X + Y equals Z.
Student_Name has entered X and Y. X * Y equals Z.
```

Where `Student_Name`, `X` and `Y` are the details entered by the user and `Z` is the result of the calculations.

Include some relevant comments in your code.

CW1SOB3 – A C program with user input and control structure

Write a **C program** that takes an exam score which is represented by an integer and provide a string result which is the classification. Include the score in the output.

The classification is as follows:

0 - 39 = FAIL
40 – 49 = THIRD
50 - 59 = LOWER SECOND
60 – 69 = UPPER SECOND
70 – 100 = FIRST

Example Input: 55

Example Output: Your exam score is 55 which is a LOWER SECOND.

The program should continue to prompt for grade input until the user enters -1.

Include some relevant comments in your code.

CW1SOB4 – A C program password entry system

Write a **C programme** that does the following:

1. Prompt the user for a whole number X.
2. Pass X to a method/function called printName.
3. The printName method/function should prompt the user for their name (userName) and then print that name X times.
4. Following this, the program should invoke a separate method/function called "passwordEntry". The "passwordEntry" method/function should prompt the user, by their provided name, to enter a password. The password will be checked against a sentinel value (determined by you – a word or a number).
5. The user must enter the correct password within 3 attempts or they will be locked out of the program.
6. The program should adequately advise the user of all instructions and warnings throughout.
7. The final output should look like the following:

   Success
   Well done userName, you got the password right in X attempts!

   Failure
   Bad luck userName, you took X attempts, you are locked out!

8. Include some relevant comments in your code.

CW1SOB5 – An advanced C program using pointers

Write a **C program** to read two numbers from the user and find the Sum, Difference, Product and Quotient. **You must use pointers** for the two numbers that were entered by the user. Print the memory address of each number **before** the calculations and **after** the calculations.

Don't forget to include comments in your code.

CW1SOB6 – A Batch Script which creates files and folders

Write a **Batch Script** that will create 3 **empty** text files and 3 folders. Each folder should contain one of the created text files. All 6 items should be in a folder named "The_Bag". Once completed, the following message should be displayed:

"The 6 items are in The Bag!".

Include some relevant comments in the script.

CW1SOB7 – A command line and Batch Script exercise

1. Using the Windows Command Line create a folder on your desktop called "CST1500", inside that folder create three more folders called "C_Programs", "Scripts" and "Docs".

2. Using the Windows Command Line, copy the FOUR C programs that you have written for this coursework into the "C_Programs" folder. Then list all of the files in the "C_Programs" folder.

3. Take a screenshot of the CMD window which will contain everything you have done so far, and paste it into a Word document (or equivalent). Caption the screenshot:

   "Fig. 1: Creating folders and copying files in the CMD window.".

4. Save the document into the "Docs" folder and name it "Screenshots".

5. Write a **Batch Script** which will list all C programs that exist in the "CST1500" directory/folder and all of its subdirectories/subfolders. Ideally, the output should be:

   The following are C programs:
   C:\Users\MyPC\MyFolder\CST1500\C_Programs\CW1SOB1.c
   C:\Users\MyPC\MyFolder\CST1500\C_Programs\CW1SOB2.c
   C:\Users\MyPC\MyFolder\CST1500\C_Programs\CW1SOB3.c
   C:\Users\MyPC\MyFolder\CST1500\C_Programs\CW1SOB4.c

   NOTE: The highlighted parts of the output should be identical in your output.

6. Take a screenshot of the CMD with the above output and paste it into the "Screenshots" document in the "Docs" folder. Caption the screenshot:

   "Fig. 2: Result of batch script listing all C programs.".

7. Move the CW1SOB7.bat file into the "Scripts" folder. This can be dragged and dropped or moved in any other way, as long as the batch file ends up in the "Scripts" folder.

8. Include some relevant comments in the script.

CW1SOB8 – A Batch Script to create and populate a text file

Write a **Batch Script** that will do the following:

1. Create a folder called “myBatchScript”.

2. Create a .txt file called “results.txt”.

3. Write the following, exactly as it appears here, to the “results.txt” file:

   Knock, knock.
   (Who’s there?)
   Recursion.
   (Recursion who?)
   Knock Knock…

4. Copy the “results.txt” file to the “myBatchScript” folder.

5. Delete the original “results.txt” file.

6. Print the message “Task Complete!”.

7. Include some relevant comments in the script.

CW1SOB9 – A Bash Script exercise

Write a **Bash Script** that will take an existing directory as an input, the script must create three folders within that path called “a”, “b” and “c” and perform the following tasks:

Task 1: Store your MDX username (on the back of your ID card) in a text file called “userName.txt” in folder “a”.

Task 2: Store the name of your favourite movie in a text file called “myFavMovie.txt” in folder “b”.

Task 3: Move folders “a” and “b” with their contents to folder “c”.

Include some relevant comments in the script.

CW1SOB10 – A Bash Script to perform admin tasks

1. Create a **Bash Script** that will take 3 parameters.

   The first parameter will be defining a path, if it doesn't exist it should be created, the script should inform the user about it.

   The second parameter will be an integer number x where x > 0 and x <= 20.

   The third parameter will be an integer number y where y > 0 and y <= 20.

2. The program should create x number of folders, the folders should be named with ascending integer numbers starting at 1.

3. Finally, the program should create y number of .txt files within each of the newly created folders, the files should be named with ascending integer numbers starting at 1.

4. Take a screenshot of the file structure and paste it into a document called "CW1SOB10_Screenshot". Caption the screenshot:

   "Fig. 1: Creating folders and files with Bash.".

   and save it in the "CourseworkOne" folder.

5. Include some relevant comments in the script.